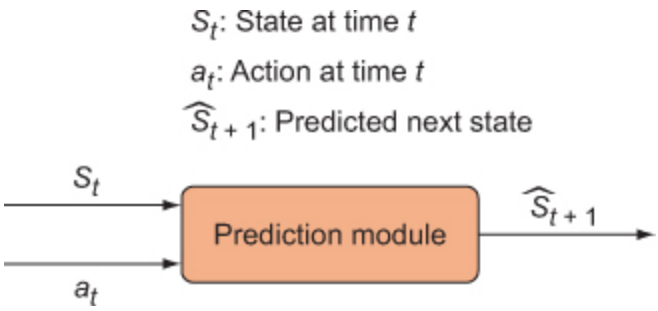
some papers require advanced mathematics, and they are outside the scope of this book, but many of the biggest papers in the field require only some basic calculus, algebra, and linear algebra—things that you probably know if you have made it this far. The only real barrier is getting past the mathematical notation, which we hope to make easier here. We want to teach you how to fish rather than just giving you fish, as the saying goes.

8.2. Inverse dynamics prediction

We've described how we could use the prediction error as a curiosity signal. The prediction error module from the last section is implemented as a function, $f:(S_t,a_t) \rightarrow \hat{S}_{t+1}$, that takes a state and the action taken and returns the predicted next state (figure 8.6). It is predicting the future (forward) state of the environment, so we call it the *forward-prediction model*.

Figure 8.6. Diagram of the forward-prediction module function, $f:(S_t,a_t) \rightarrow \check{S}_{t+1}$, which maps a current state and action to a predicted next state.



Remember, we want to only predict aspects of the state that actually matter, not parts that are trivial or noise. The way we build in the "doesn't matter" constraint to the prediction model is to add another model called an *inverse model*, $g:(S_t,S_{t+1}) \rightarrow \hat{a}_t$. This is a function, g, that takes a state and the next state, and then returns a prediction for which action was taken that led to the transition from s_t to s_{t+1}, as shown in figure 8.7.

Figure 8.7. The inverse model takes two consecutive states and tries to predict which action was taken.